



EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	% / CGPA	Year
MS (Research)	IIT Madras	8.2	2027*
B.E. in Mechanical Engg.	A.C. Patil College of Engg.	9.04	2022
Diploma in Mechanical Engg.	Bharati Vidyapeeth Inst. of Technology	80.65 %	2019
Class X (SSC)	Sandesh Vidyalyaya	88.20%	2016

Scholastic Achievements : AIR 547 in GATE'24 (XE) out of 11000+ aspirants, Awarded a full-fee undergraduate scholarship by the **Rotary Club of Bombay**, for up to six years.

RESEARCH EXPERIENCE

TOPIC : Interfacial Mixed-Mode Fracture of Hyperelastic Thin Films on Curved Substrates

MS Research (Dec 2025 – Present) <i>Guide:</i> Dr. Shuvrangs Das	- Built Nonlinear finite element models for Mixed mode thin film peeling on curved surfaces, incorporating hyperelastic material modelling of the thin film, using Abaqus based Cohesive zone modelling (CZM). - Built Python automated parametric geometric models of convex and concave substrates in Abaqus, capturing varying curvature signs and radii to support systematic peeling-response studies across substrate geometries. - Designed and fabricated custom gigs, fixtures and substrates using SolidWorks to enable repeatable, cost-effective experimental validation of displacement controlled peeling. - Achieved ~96% agreement between measured and predicted deformation trends by calibrating a GoPro camera (checkerboard method) and building a MATLAB-Image processing pipeline to track markers and extract required data. - Validated simulation accuracy to within ~10% normalized error against experiments by running parametric studies on CZM damage parameters and benchmarking predicted force-displacement response against measured data. - Quantified Rate-dependent/viscoelastic debonding behavior across multiple peeling rates by designing and running displacement-controlled peeling experiments and analyzing force-displacement response. - Characterized the influence of substrate geometry, curvature sign, and peeling direction on mixed-mode interfacial debonding, providing mechanistic insights relevant to adhesion and reliability challenges in curved, flexible thin film electronic systems.
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PROFESSIONAL EXPERIENCE

Solar Design Engineer (July 2022 – Feb 2023) <i>"Green world renewable energy LLC."</i>	• Designed residential and utility solar PV¹ permit designs across the USA, preparing AHJ² -compliant plansets based on customer and site requirements. • Prepare roof/site layouts, CAD³ drawings (2D), Electric Line diagrams (ELD), SLD⁴, structural calculations, equipment datasheets, and supporting documentation in accordance with NEC⁵ and jurisdictional requirements. • Analyzed satellite imagery and client-provided site photographs to determine solar array layouts, equipment placement, and site constraints. • Selected required system components and additional devices based on project specifications and code requirements.
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COURSE PROJECTS

Composite Structures (Jan '25-May '25)	• Static Analysis of Carbon Fibre Reinforced Polymer(CFRP)-Strengthened Reinforced Concrete Beam in Abaqus - Developed and analysed an Abaqus finite element model of CFRP-strengthened reinforced concrete beams, modelling concrete, steel reinforcement, and CFRP laminates to assess their influence on stress-strain behaviour, deformation response, and structural performance under static loading.
Advanced Solid Mechanics (July '25-Nov '25)	• Mixed mode fracture on curved substrate for Hyperelastic materials - Implemented and validated the analytical formulation of a V-shaped double peeling model in MATLAB by reproducing published results and extending the methodology to a related interfacial peeling problem.
Nonlinear finite element analysis (Jan '26-May '26)	• Finite deformation framework for Neo-Hookean materials - Implemented a Python-based nonlinear finite deformation framework for Neo-Hookean hyperelasticity, evaluating kinematics, strain energy, and stress evolution under multiple deformation modes.

OTHER PROJECTS

Interfacial Fracture Mechanics	• Parametric and Sensitivity Analysis of Abaqus-CZM 90° Peel Models  - Performed model verification, benchmark validation against published literature, and sensitivity analysis to identify robust modelling practices, achieve numerically stable solutions, and establish reliable simulation frameworks for interfacial fracture and peeling mechanics studies.
Image Analysis-MATLAB pipeline	• Camera Calibration and Image based Measurement using MATLAB  - Built an end to end MATLAB pipeline converting raw experimental video into quantitative displacement data with checkerboard camera calibration, automated time interval frame extraction, marker based feature tracking and image to world coordinate transformation. - Methodology developed makes the pipeline directly reusable for any camera-based displacement/deformation measurement such as tensile, bending, shear or full-field tracking.

Python based T-S law simulator	• CZM Traction-Separation Law Visualizer 			
	- Clarified how damage evolves differently across three CZM formulations Exponential, Bilinear and Trapezoidal , measured by animated traction-separation and damage curves under cyclic, irreversible loading , by implementing standalone python based simulations for each traction-separation law.			
Undergraduate (BE) Project	• Fabrication of Low-Cost Organic Waste Shredder			
	- Designed, fabricated, and tested a low-cost organic waste shredding machine for efficient natural fertilizer and compost production to reduce processing costs and improve accessibility for small-scale and economically disadvantaged farmers.			
TECHNICAL SKILLS				
Programming	Python (NumPy, Matplotlib, OpenCV), Dolfin, MATLAB, Mathematica, Abaqus scripting			
Computational Mechanics and Modelling	Hyperelastic material modelling, Non-linear finite element analysis, Computational mechanics, Parametric and Sensitivity analysis, Interfacial fracture mechanics			
Software and Simulation tools	Abaqus, Solidworks, AutoCAD, ParaView, Gmsh			
Experimental Techniques	Test Gigs/Fixture design, Image Analysis, Camera Calibration, Visco-elastic response analyses, Feature tracking			
Writing/ Designing tools	LATEX, Inkscape, Adobe photoshop			
RELEVANT COURSES				
Composite structures	Continuum Methods in Mechanics	Finite Element Analysis	Nonlinear Finite Element Analysis	Advanced Solid Mechanics
Strength of materials	Theory of machines	Vibration	Machine Design	Metrology
Manufacturing Science	Operations Research	Thermodynamics	Heat and Mass transfer	Fluid Mechanics
CONFERENCE PRESENTATION				
INCAM 2026	Effects of Curvature Sign on the Peeling of Thin Films From Curved Substrates			July'26
	IIT Kanpur			Indian Conference on Applied Mechanics (INCAM 2026)
	- Presented research on mixed-mode peeling and interfacial failure of thin films on curved substrates, highlighting its potential implications for Semiconductor packaging and Flexible electronics .			
WORKSHOPS				
PFEAA'25	PRACTICAL FINITE ELEMENT ANALYSIS WITH ANSYS NIT Calicut			June'25
	- Developed practical proficiency in finite element modelling, simulation workflows, and result interpretation for structural engineering problems.			
ML for Scientific Computing	MACHINE LEARNING FOR SCIENTIFIC COMPUTING IIT Bombay			July'26
	- Studied numerical methods, finite element formulations, and discretization techniques for solving ordinary and partial differential equations in engineering systems. - Gained exposure to supervised machine learning, artificial neural networks, and physics-informed machine learning for data-driven modelling and differential equation-based scientific computing.			
ONLINE COURSE CERTIFICATIONS				
ML for engg. and science application(NPTEL)		Abaqus Python scripting(Udemy)		Computer Vision Onramp(Mathworks)
Adhesive joints and Composite material(Udemy)				
POSITIONS OF RESPONSIBILITY				
AIRSS'25	• Coordinator, Design Team			All India Research Scholar's Summit' 2025
	- Coordinated design, publicity, and presentation activities for AIRSS 2025 at IIT Madras, developing promotional and event communication materials, anchoring technical presentation sessions, and facilitating the participation of 600+ research scholars from across India to ensure seamless summit operations.			
Discovering Creativity	• Teaching Assistant			
	- Supported course delivery for 50+ students by facilitating classroom activities, grading assignments and quizzes, and assisting faculty with assessments and academic coordination.			
Applied Mechanics Lab	• Teaching Assistant			
	- Conducted tensile testing sessions for 200+ students , demonstrating UTM ⁶ operation and stress–strain analysis.			
HOBBIES				
Trekking and Nature exploration, Cycling, Badminton, Pencil sketching and painting, Music, Meditation				

¹PV- Photovoltaic; ²AHJ- Authorities having jurisdiction; ³CAD- Computer aided drawings; ⁴SLD- Single line diagrams; ⁵NEC- National electric code; ⁶UTM- Universal tensile machine